

MANILA NINOY AQUINO INTL, Philippines

WMO: 984290

Lat: 14.509N

Lon: 121.020E

Elev: 23

StdP: 101.05

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	22.8	16.7	11.9	26.9	17.3	12.4	26.8	8.4	28.7	7.4	28.8	1.2	70	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.3	35.1	25.9	34.3	25.8	33.9	25.8	28.5	30.6	28.1	30.2	27.8	30.0	4.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.4	29.2	27.8	23.9	29.1	27.2	23.0	28.8	92.5	30.8	90.7	30.2	89.3	30.1	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.4	6.4	DB	20.3	36.8	1.1	0.8	19.5	37.4	18.9	37.8	18.3	38.3	17.5	38.9	(4)
			WB	17.7	29.8	1.1	0.7	16.9	30.3	16.3	30.7	15.7	31.1	14.9	31.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	26.7	27.2	28.4	30.0	30.1	29.2	28.2	28.0	28.0	28.1	28.0	27.1	(6)
(7)		DBStd	1.57	1.13	1.16	1.18	1.18	1.44	1.35	1.29	1.13	1.07	1.03	1.08	1.25	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6659	517	482	571	599	623	575	564	557	539	562	539	532	(10)
(11)		CDD18.3	3617	258	248	312	349	365	325	306	298	289	304	289	273	(11)
(12)		CDH23.3	40427	2329	2431	3561	4559	4758	3963	3353	3248	3108	3372	3113	2632	(12)
(13)		CDH26.7	15394	633	781	1416	2252	2362	1712	1178	1065	1008	1168	1065	755	(13)
(14)	Wind	WSAvg	2.7	2.6	3.0	3.2	3.2	2.9	2.7	2.8	3.1	2.7	2.3	2.3	2.3	(14)
(15)	Precipitation	PrecAvg	2200	26	24	27	25	148	220	415	463	389	228	122	110	(15)
(16)		PrecMax	3377	102	128	122	161	422	557	1235	1113	898	474	278	395	(16)
(17)		PrecMin	1442	0	0	0	0	8	58	107	228	155	53	0	0	(17)
(18)		PrecStd	481	27	34	32	36	98	111	234	213	145	101	79	90	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	33.1	34.2	36.0	36.2	35.2	34.1	33.1	33.2	33.2	33.2	32.9	(19)
(20)		0.4%	MCWB	24.6	24.5	25.1	25.4	25.8	26.7	26.5	26.9	26.6	26.1	25.6	25.0	(20)
(21)		2%	DB	31.1	32.1	33.4	35.1	35.2	34.1	33.0	32.0	32.1	32.4	32.2	31.9	(21)
(22)		2%	MCWB	24.3	24.4	24.9	25.5	26.0	26.7	26.6	26.6	26.7	26.0	25.5	24.9	(22)
(23)		5%	DB	30.4	31.3	32.9	34.2	34.6	33.1	32.0	31.2	31.2	31.8	31.9	31.0	(23)
(24)		5%	MCWB	24.0	24.1	24.8	25.3	26.0	26.6	26.5	26.6	26.8	26.0	25.5	24.9	(24)
(25)		10%	DB	29.8	30.7	32.0	33.7	33.8	32.2	31.0	30.8	30.8	31.0	31.0	30.1	(25)
(26)		10%	MCWB	23.9	23.9	24.4	25.2	26.1	26.6	26.6	26.6	26.8	25.9	25.5	24.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	26.6	26.9	28.1	28.7	28.9	28.8	28.7	28.8	28.6	28.1	27.2	(27)
(28)		0.4%	MCDWB	28.9	29.8	31.0	30.8	31.5	31.3	30.4	30.3	30.4	30.4	30.1	28.9	(28)
(29)		2%	WB	25.9	25.8	26.2	27.2	28.0	28.2	28.1	28.1	28.2	27.9	27.2	26.6	(29)
(30)		2%	MCDWB	28.6	29.3	30.6	31.2	31.0	30.6	30.1	29.9	30.0	29.6	29.7	28.7	(30)
(31)		5%	WB	25.3	25.3	25.8	26.6	27.5	27.7	27.6	27.6	27.7	27.4	26.7	26.1	(31)
(32)		5%	MCDWB	28.3	29.1	30.3	31.2	30.9	30.4	29.8	29.6	29.6	29.3	29.3	28.5	(32)
(33)		10%	WB	24.8	24.7	25.4	26.2	27.1	27.3	27.1	27.1	27.2	27.0	26.2	25.7	(33)
(34)		10%	MCDWB	28.0	28.7	29.8	31.1	30.9	30.3	29.6	29.3	29.2	29.1	28.8	28.2	(34)
(35)	Mean Daily Temperature Range	MDBR	6.8	7.3	7.8	8.0	7.3	6.5	5.9	5.4	5.6	6.0	6.5	6.2	(35)	
(36)		5% DB	MCDBR	7.4	8.1	8.4	8.5	8.3	7.5	7.0	6.3	6.5	7.1	7.4	7.2	(36)
(37)		5% DB	MCWBR	3.1	3.1	3.2	3.0	2.9	2.8	3.0	2.9	3.0	3.0	2.9	2.9	(37)
(38)		5% WB	MCDBR	6.5	7.1	7.6	7.8	7.2	6.6	6.0	5.6	5.9	6.0	6.5	6.1	(38)
(39)		5% WB	MCWBR	3.3	3.2	3.3	3.2	3.1	3.0	3.2	3.1	3.2	3.2	3.1	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.421	0.419	0.450	0.474	0.474	0.467	0.463	0.470	0.466	0.477	0.439	0.441	(40)	
(41)		taud	2.416	2.409	2.325	2.262	2.292	2.327	2.339	2.308	2.316	2.266	2.387	2.367	(41)	
(42)		Ebn at Noon	874	892	870	843	830	827	832	835	843	826	849	843	(42)	
(43)		Edh at Noon	114	119	133	141	135	128	127	133	132	136	116	117	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.20	4.97	5.62	6.32	5.72	5.13	4.52	4.37	4.62	4.44	4.17	3.73	(44)	
(45)		RadStd	0.26	0.43	0.39	0.34	0.51	0.51	0.57	0.45	0.46	0.36	0.37	0.33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.42	+0.82	N/A	+0.58	N/A	N/A	N/A	N/A	+183	+183	(46)	
(47)	Regional (16 neighbors)	+0.32	+0.47	N/A	+0.45	+0.40	+0.54	N/A	N/A	+158	+158	(47)	

Nomenclature: See separate page